



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	CASE-df005-2017-android
Operator:	Floris
Analysis Started:	2026-06-17 12:22:46
Analysis Completed:	2026-06-17 12:25:43
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df005\2017\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df005\2017\dfdof_output\CASE-df005-2017-android
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df005_external.001	drone_sd	physical	14.9 GB	2026-06-17 12:23:33
DF005_flight.zip	controller_android	logical	657.1 MB	2026-06-17 12:23:35

Evidence Source Path

C:\Users\Floris\Documents\ddj_drones\test_phantom_4\df005\2017\android

Cryptographic Hashes

df005_external.001		drone_sd	physical
SHA-256:	15f4b7897912ab70c151f23ff4050677e622b2d84412c7d6c6616952d9f8f7e8		
SHA-1:	dc26935a64954a24a41776cba813b4c1dea80017		

DF005_flight.zip		controller_android	logical
SHA-256:	89b77f96d1d76dcbdd18459f07e20bcacfa23fc6ae6984c4ec4bc005bb7c9308		
SHA-1:	820c2cd5a003681f0c24d4dcdeebbb4d68fd4968		

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for DF005_flight.zip
2. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for DF005_flight.zip
3. [p2 - controller ios]: source evidence not found
4. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
5. [p3 - controller android - drone logs]: no artefacts found
6. [p3 - controller android - images]: no artefacts found
7. [p3 - controller ios]: source evidence not found
8. [p3 - drone flight storage]: source evidence not found
9. [p4 - controller ios]: source evidence not found
10. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	26	26	[OK]
fls	1	1	[OK]
icat	25	25	[OK]
mmls	1	1	[OK]
parser_android	1	1	[OK]
txtlogtocsvtool	2	2	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	6 / 7
Flights With Primary Correlation	0 / 2
Tools Succeeded	6 / 6

3. Device and Account Identification

Field	Value	Confidence	Sources
Drone Serial Number	07JDE5300200GN	[K] Controller-only	3
Drone Model	P4 Professional	[K] Controller-only	2
DJI Application Version	4.1.2	[K] Controller-only	2
Account E-Mail	droneforensics@vtointc.com	[K] Controller-only	1
Product Type	24	[K] Controller-only	1
Last Country Code	US	[K] Controller-only	1
Aircraft Model Code	wm331	[K] Controller-only	1
Controller Serial Number	0BDDE5900100GD	[K] Controller-only	1
Last Known Latitude	39.96117388555166	[K] Controller-only	1
Last Known Longitude	-106.21649606984288	[K] Controller-only	1
Account UID	879825329191059456	[K] Controller-only	2
Account Nickname	dronefo...	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 6911afe465459133dec1f6dec70ca16ed4454fa5caa9699f4398b457dc895399
Flight Record Name: DJIFlightRecord_2017-06-29_[11-15-17].csv
Start: 2017-06-29T17:15:17.833000+00:00
End: 2017-06-29T17:16:33.564000+00:00
Duration: 75.7 s

Flight Metrics

Peak Height: 2.9 m
Distance Travelled: 5.65 m
Video Recording: No
Photo Capture: No
Photos Taken: 0
Recording Duration: 0.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

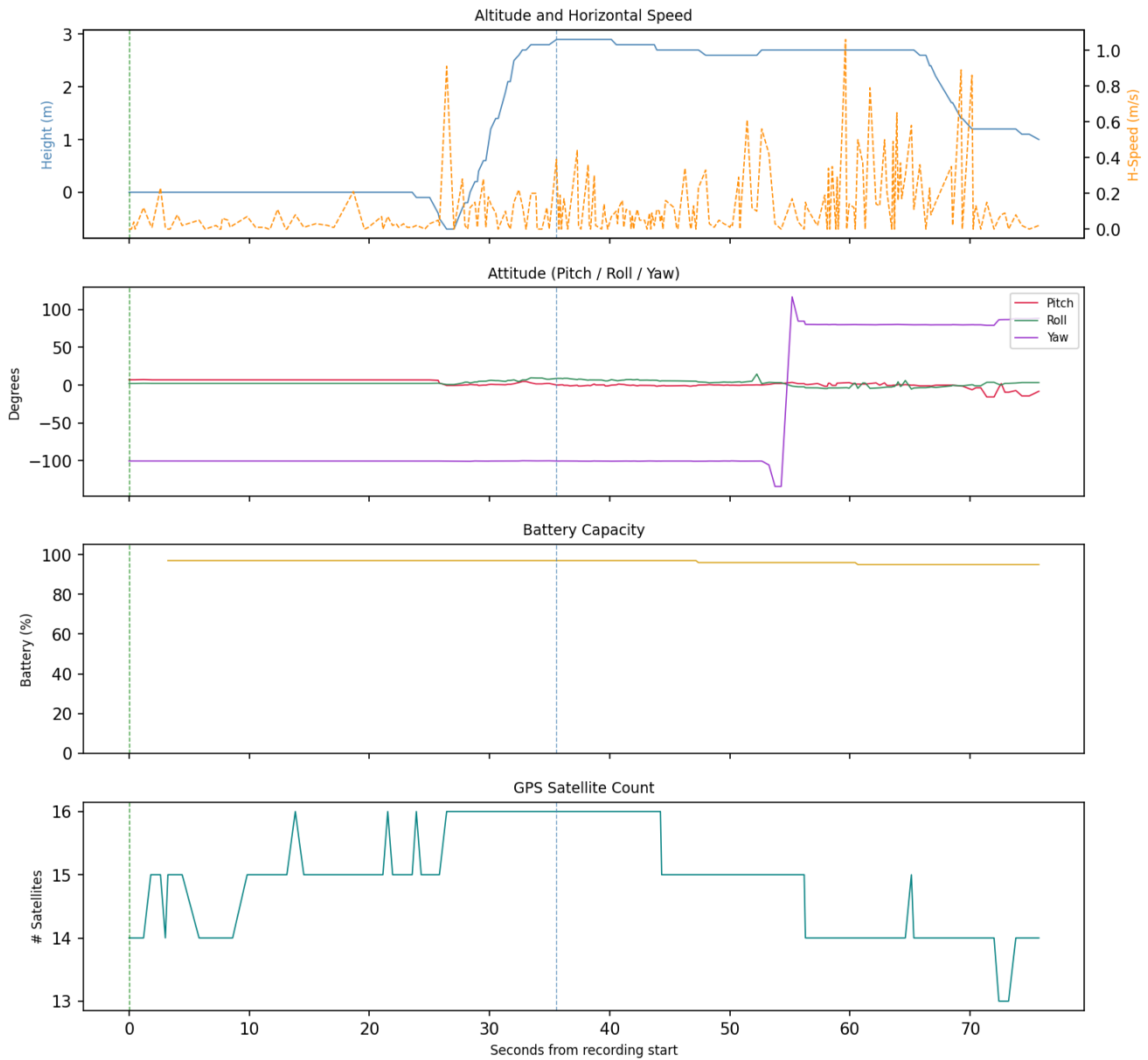
1. EngineStart
2. AssistedTakeoff
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
2017-06-29 17:15:17	ctrl_and:fr (v4.1.2)	Log started	SN: 07JDE5300200GN; App: 4.1.2; P4 Professional
2017-06-29 17:15:17	ctrl_and:fr (v4.1.2)	Motor turned on	on
2017-06-29 17:15:53	ctrl_and:fr (v4.1.2)	Reached peak height	2.9 m
2017-06-29 17:16:33	ctrl_and:fr (v4.1.2)	Log ended	Dist home: 1.5 m; Height: 1.0 m; Rec: 0.0 s

Flight Record - Sensor Telemetry

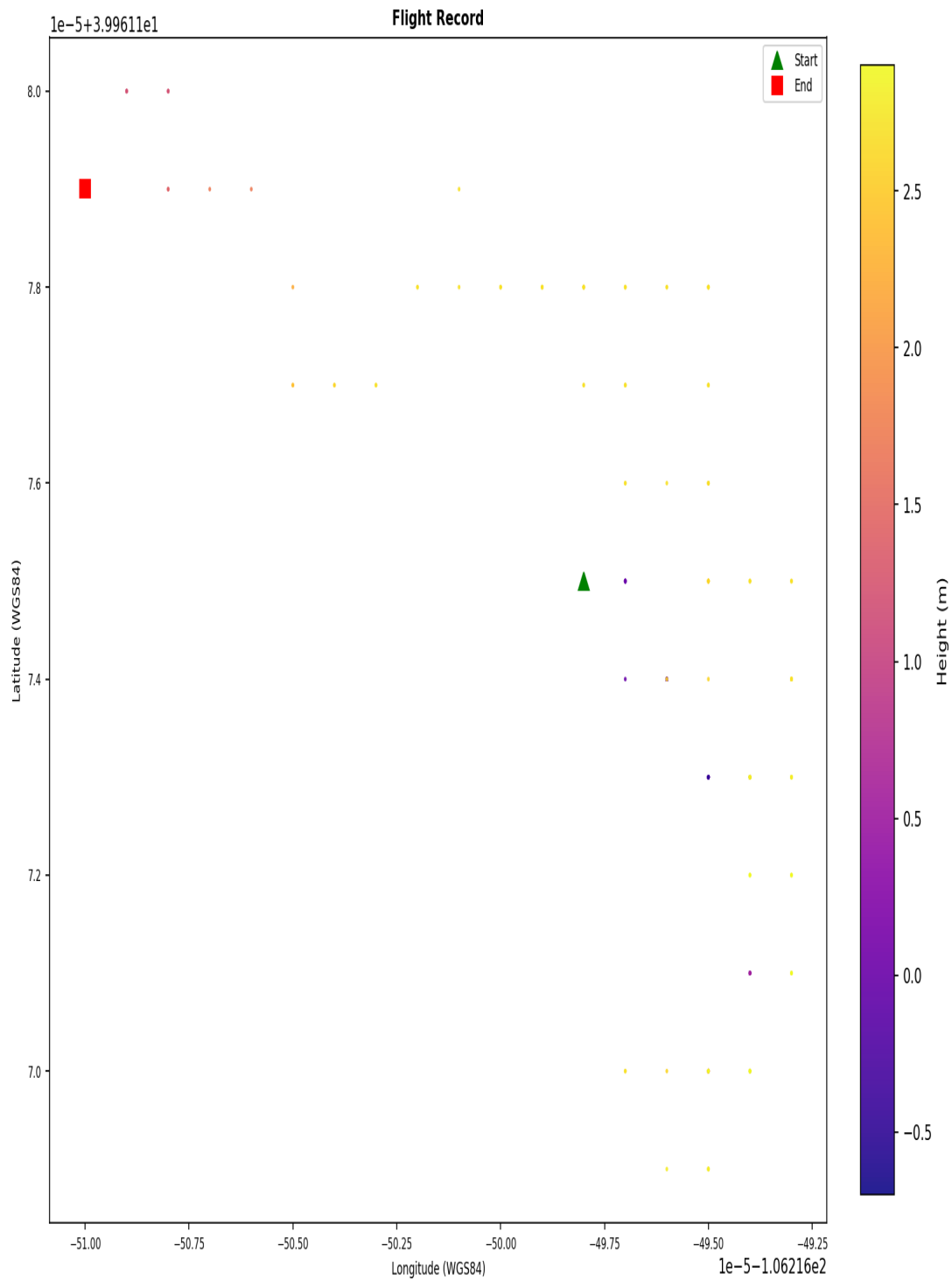
Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_01



Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 0f734a559c7d1a1b69c5b0436dcaa91c3cf3ac33c4f9144c96808e60e12d7f14
Flight Record Name: DJIFlightRecord_2017-06-29_[11-20-32].csv
Start: 2017-06-29T17:20:32.085000+00:00
End: 2017-06-29T17:36:14.407000+00:00
Duration: 942.3 s

Flight Metrics

Peak Height: 43.6 m
Distance Travelled: 2759.66 m
Video Recording: No
Photo Capture: No
Photos Taken: 9
Recording Duration: 0.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. AutoLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-06-29 17:20:32	ctrl_and:fr (v4.1.2)	Log started	SN: 07JDE5300200GN; App: 4.1.2; P4 Professional
2017-06-29 17:20:32	ctrl_and:fr (v4.1.2)	Motor turned on	on
2017-06-29 17:29:28	ctrl_and:fr (v4.1.2)	Reached peak height	43.6 m
2017-06-29 17:36:14	ctrl_and:fr (v4.1.2)	Log ended	Dist home: 2.0 m; Height: 4.4 m; Rec: 0.0 s

Correlated Artefacts

Possibly correlated (indirect indicators):

controller_android:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:9a5661921fe118d55c89cfbd0abd9f26d668d5d3208f4c4437fe1a32551f9448	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:7b8d41c1775714af2eadff7cf56c9acbf4b0d80a1c53867d061be7e25eea680	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:83d34e5880b8be3bde56bd87afebd103592a5e5994304a995d036034962a8272	
drone_sd:videos	GPS within flight bounding box (lat 39.9585, lon -106.2203)
p5:84b82a66fa36d126ed580f2d91bc47d6f4787b93bec75585ba23dfb4068eb19	
drone_sd:videos	GPS within flight bounding box (lat 39.9585, lon -106.2203)
p5:624ccfe3d1f63aad44f49156a8f89326e679a81087999ebb77f6707815e350bb	
drone_sd:videos	GPS within flight bounding box (lat 39.9611, lon -106.2166)
p5:07193b3b3f1f024c7b8e66cbe104bd4106db3269a6f278245056cb63ec837132	
drone_sd:videos	GPS within flight bounding box (lat 39.9611, lon -106.2166)
p5:d1c7d4c9532dd338776e728447b3f9ac9637aa0df2403597cebae87a12fce7ae	
drone_sd:videos	GPS within flight bounding box (lat 39.9582, lon -106.2172)

p5:c29a9498a07292ad1a133d77584e928ee5a05147cfd4d3b770f4d7c6ada17406

drone_sd:videos	GPS within flight bounding box (lat 39.9582, lon -106.2172)
-----------------	---

p5:14fbc8239759ab3953aca18c3b7ef9beccc96ee09c67943bfe7483dbfff99068

drone_sd:videos	GPS within flight bounding box (lat 39.9588, lon -106.2202)
-----------------	---

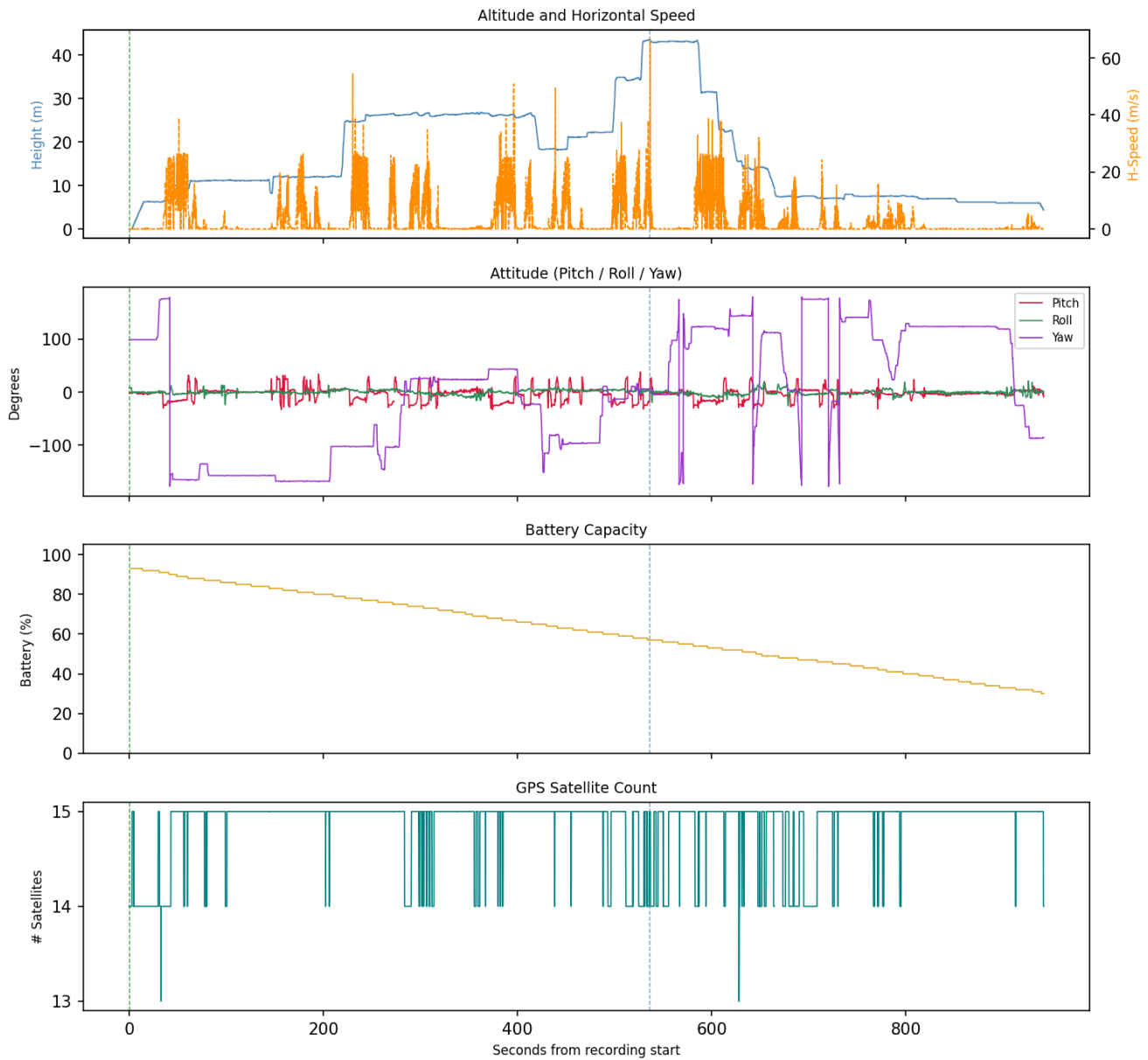
p5:7eecdb89bdc561442c19cd3b6c462e508398bb945657f0acb4fb3a3b29b18f55

drone_sd:videos	GPS within flight bounding box (lat 39.9588, lon -106.2202)
-----------------	---

p5:bb2ec42222e8a87726519e1846155bc5937aa0b31f6a797cd97e71376e65aa45

Flight Record - Sensor Telemetry

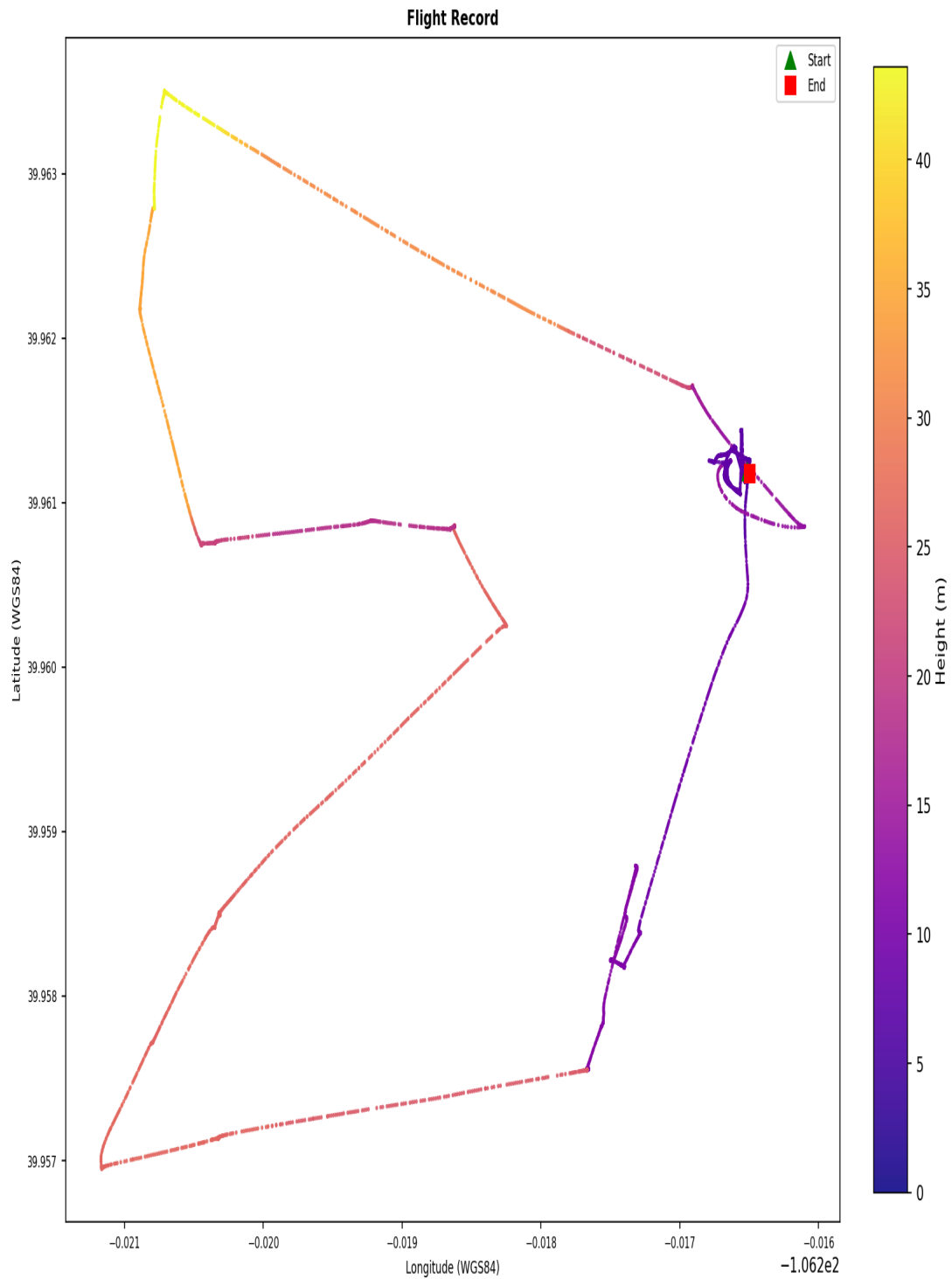
Flight Record Telemetry - flight_02



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_02



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	2	7	0	35	2	0	2
drone_sd	0	0	0	0	0	15	10

Data Quality Notes per Category

Flight Logs (35 files):

- formats: 3 bracketed_log, 1 heading_log, 31 unknown

Flight Records (2 files):

- 1 file(s) with row anomalies
- 2 file(s) with empty columns
- 2 file(s) with constant-value columns

Images (15 files):

- 15 file(s) contain(s) EXIF GPS coordinates
- 15 file(s) missing (normalisable) EXIF timestamp

Videos (12 files):

- 11 file(s) contain(s) EXIF GPS coordinates
- 11 file(s) missing (normalisable) EXIF timestamp
- 1 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Flight_logs: log files present in the following format(s): 3 bracketed_log, 1 heading_log, 31 unknown.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 2 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 2 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android - device and backup info]: DeviceInfo.xml not found for DF005_flight.zip
2. [p2 - controller android - device and backup info]: ApplicationInfo.xml not found for DF005_flight.zip
3. [p2 - controller ios]: source evidence not found
4. [p3 - controller android]: installed DJI apps not found in P2 output; falling back to config-defined bundle IDs for extraction
5. [p3 - controller android - drone logs]: no artefacts found
6. [p3 - controller android - images]: no artefacts found
7. [p3 - controller ios]: source evidence not found
8. [p3 - drone flight storage]: source evidence not found
9. [p4 - controller ios]: source evidence not found
10. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Flight Logs

29-06-2017

- format: heading_log | entries: 27

log-2017-06-28.txt

- format: unknown | entries: 0

log-2017-06-28_1.txt

- format: unknown | entries: 0

log-2017-06-28_10.txt

- format: unknown | entries: 0

log-2017-06-28_11.txt

- format: unknown | entries: 0

log-2017-06-28_12.txt

- format: unknown | entries: 0

log-2017-06-28_13.txt

- format: unknown | entries: 0

log-2017-06-28_14.txt

- format: unknown | entries: 0

log-2017-06-28_15.txt

- format: unknown | entries: 0

log-2017-06-28_16.txt

- format: unknown | entries: 0

log-2017-06-28_17.txt

- format: unknown | entries: 0

log-2017-06-28_18.txt

- format: unknown | entries: 0

log-2017-06-28_2.txt

- format: unknown | entries: 0

log-2017-06-28_3.txt

- format: unknown | entries: 0

log-2017-06-28_4.txt

- format: unknown | entries: 0

log-2017-06-28_5.txt

- format: unknown | entries: 0

log-2017-06-28_6.txt

- format: unknown | entries: 0

log-2017-06-28_7.txt

- format: unknown | entries: 0

log-2017-06-28_8.txt

- format: unknown | entries: 0

log-2017-06-28_9.txt

- format: bracketed_log | entries: 4

log-2017-06-29.txt

- format: unknown | entries: 0

log-2017-06-29_1.txt

- format: unknown | entries: 0

log-2017-06-29_10.txt

- format: bracketed_log | entries: 1

log-2017-06-29_11.txt

- format: bracketed_log | entries: 2

log-2017-06-29_12.txt

- format: unknown | entries: 0

log-2017-06-29_13.txt

- format: unknown | entries: 0

log-2017-06-29_14.txt

- format: unknown | entries: 0

log-2017-06-29_2.txt

- format: unknown | entries: 0

log-2017-06-29_3.txt

- format: unknown | entries: 0

log-2017-06-29_4.txt

- format: unknown | entries: 0

log-2017-06-29_5.txt

- format: unknown | entries: 0

log-2017-06-29_6.txt

- format: unknown | entries: 0

log-2017-06-29_7.txt

- format: unknown | entries: 0

log-2017-06-29_8.txt

- format: unknown | entries: 0

log-2017-06-29_9.txt

- format: unknown | entries: 0

Flight Records

norm_DJIFlightRecord_2017-06-29_[11-15-17].csv

- altitude_negative: 7 row(s) affected
- contains_no_value: 58 column(s) - APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CAMERA_INFO.cameraType, CAMERA_INFO.connectState (+53 more)
- contains_constant_value: 157 column(s) - APP_GPS.accuracy, APP_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.errorType, CENTER_BATTERY.fullCapacity [mAh] (+152 more)

norm_DJIFlightRecord_2017-06-29_[11-20-32].csv

- contains_no_value: 52 column(s) - APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CAMERA_INFO.cameraType, CAMERA_INFO.connectState (+47 more)
- contains_constant_value: 159 column(s) - APP_GPS.accuracy, CENTER_BATTERY.connStatus, CENTER_BATTERY.errorType, CENTER_BATTERY.fullCapacity [mAh], CENTER_BATTERY.life (+154 more)

Images

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0004.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0006.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0007.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0008.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0010.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0011.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0013.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0014.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0015.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0016.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0017.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0019.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0020.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

Videos

2017_06_29_11_20_42.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0001.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0012_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0012.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0018_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0018.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no installed DJI applications found - Device and account information.
2. Note: no controller device name found - Device and account information.
3. Note: no controller device found - Device and account information.
4. Note: no firmware version found - Device and account information.
5. Note: no device uuid found - Device and account information.
6. 2 flight record(s) could not be correlated to a drone log (flight_01, flight_02). Independent corroboration is absent for these flight(s).
7. 7 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

dji.go.v4.xml	Account Data	controller_android	extract_logical
SHA-256:	c81b32d27001231dc339d0e339d88cd80fee75c6e5bc3d39c96cd7fa333e25d5		
SHA-1:	3b249dac1a2a81f63cedd0da325c48eec0574f3f		

dji.pilot.xml	Account Data	controller_android	extract_logical
SHA-256:	ead602dbe25be4ebcf1637f281ad08ad69e481beff0a81646cba50f64a0e6e64		
SHA-1:	8098e856e52a1b1b98bed64252f8223a70601fa9		

dji.db	Databases	controller_android	extract_logical
SHA-256:	6fccc7ce013a6adacdba35ce7ab0473540d3559739f73b20747cab92b4cf8467		
SHA-1:	4401de8914f4da25086b22f56b1c18a79b83a1a7		

dji_1.db	Databases	controller_android	extract_logical
SHA-256:	03893b82908e748ec88dd1fe47fe5c8ce39e3a63bea63401eda62760e2d669c1		
SHA-1:	8a131b3c180b11fd0451afae0c9ff39ae8130a17		

flysafe_polygon_1860.db	Databases	controller_android	extract_logical
SHA-256:	de615b0a8e225bc1a815b94308e7d1644b648995c43bc0af075f0990adadb90d		
SHA-1:	220f2c8924ad0b196f233dc26b4adfb30367d539		

jpush_local_notification.db	Databases	controller_android	extract_logical
SHA-256:	c785ff31316c80b481d1ba6da796f401253cd17c812b3a41988c8aa9faea89b6		
SHA-1:	3d1cb1b9a2f2d533e0a84d01c7a4e3f9c115c85d		

jpush_statistics.db	Databases	controller_android	extract_logical
SHA-256:	0940f4ef320c1660382c1e2162f533cd7e0a98e417c40d5b343f1644fd41a3fc		
SHA-1:	e0bfb97f7bc4e0cc188acac7d899b8b7f7731abd		

logdb.db	Databases	controller_android	extract_logical
SHA-256:	90b607612fb1bc79f1d604727803ddf6161d5d56ee3afbe04989a66183d938a9		
SHA-1:	32b57c3782cfdaaea4f8cba76c599ae8dcd04727		

service.db	Databases	controller_android	extract_logical
SHA-256:	2f839d8db239fbac29b0c5e4e12561a35331368506f12a70478ba75a9c16a5b5		
SHA-1:	f1db295bb6b5ba9b3522d63eb4469ccba6641acb		

29-06-2017	Flight Logs	controller_android	extract_logical
SHA-256:	e9b72431914a5f2d51696f4efb79a2d9da088c83a3e3f8a1266caa28b377caf7		
SHA-1:	4145ee2c49d430e31a023e3ac95073963150a292		

log-2017-06-28.txt	Flight Logs	controller_android	extract_logical
SHA-256:	dc7017e933051f6500549e7028683e44e04bd5e1cfac9c4414a5e5b03e15163b		
SHA-1:	6c5db04e66c041e4e84f89fe7a7771812b442639		

log-2017-06-28_1.txt	Flight Logs	controller_android	extract_logical
SHA-256:	994119fd253212cd526c86b1a3dc2d9dec3490e468797596d92c58c6f389c9a1		
SHA-1:	6af64af8647d805714c87488b5e691730ffda1e3		

log-2017-06-28_10.txt	Flight Logs	controller_android	extract_logical
SHA-256:	447bfffdb650302376af587602909749675b36a17a5d15de7107d13fcf1113dd		
SHA-1:	55fe06f5a070318efa0b7f9b34bf35aeee7b7f9d		

log-2017-06-28_11.txt	Flight Logs	controller_android	extract_logical
SHA-256:	77e4964647be992e7776f80fb5fa695e8492f476ce60ced654c71f59124074fd		
SHA-1:	982da0ca232b7f926c5cc8a9be3836bbab59d783		

log-2017-06-28_12.txt	Flight Logs	controller_android	extract_logical
SHA-256:	db0fbc11d7d6cb23fed591b0ef6742d76cd9c4b20902f80b7e3fda339482b188		
SHA-1:	041cb4ad4dcd6b428828b4fa82a490d229ceadd3		

log-2017-06-28_13.txt	Flight Logs	controller_android	extract_logical
SHA-256:	85e2125a9c837452c2b99df84e59f3dd98d35a78b5f9c5e8bacf78c13d6ed626		
SHA-1:	a4f9561d2ba2b0a33a4a64dd0e3e7ff82c0f835a		

log-2017-06-28_14.txt	Flight Logs	controller_android	extract_logical
SHA-256:	568f913ba0e7427b019b06a9becc2f7dd509c5c7b85d71bf3bc8b7820afa28c		
SHA-1:	8f3461af375c54a4c88d22adf3239e65e294f3fc		

log-2017-06-28_15.txt	Flight Logs	controller_android	extract_logical
SHA-256:	c97f0effc0c777a5668bf0a163c387b87b55cdc18450678015357da1c80ff862		
SHA-1:	b824e454c3327583742d7801f2136bbe85ab3ed7		

log-2017-06-28_16.txt	Flight Logs	controller_android	extract_logical
SHA-256:	0db2b44a5a0d900ddc39809e677dd91afd31f5cc723249f3cc626b144f743edd		
SHA-1:	9d2410c1aeddacddf759bd28d008c1e284663282		

log-2017-06-28_17.txt	Flight Logs	controller_android	extract_logical
SHA-256:	cdeb33d92d5859816e108202e558068ee93e215c1cc82f61ed9836ad8d0804c0		
SHA-1:	7ccbba75b67da006efa3248a6ff927e62d9e3996		

log-2017-06-28_18.txt	Flight Logs	controller_android	extract_logical
SHA-256:	d31b576519f896854e9cb1a80aa3630dba3ff8dc33d9cc4cb12b4924b3fa9343		
SHA-1:	74958a987691cd102aba1894b1ec0a9a774fedb0		

log-2017-06-28_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	f7f7483e0face67a7d3e3cb3eacd2968e452053d5ca4c2a474e0da39dfd392b5		
SHA-1:	cd70d20cb01906b947299143ffc530344b340d81		

log-2017-06-28_3.txt	Flight Logs	controller_android	extract_logical
SHA-256:	aee8d6f89ad95094c012990bb08046e5bd837e9b0916724c72ed73d193bc32ce		
SHA-1:	fa3d8a4812be30a188a3eeb7280c881a170fd4b9		

log-2017-06-28_4.txt	Flight Logs	controller_android	extract_logical
SHA-256:	29a8227eb4ca08794777fa322c27b777f52b4ac3995cfe4d2d2f2087e7624935		
SHA-1:	05cf377637ae2d8c2c96abd48576f4f1d7ac6880		

log-2017-06-28_5.txt	Flight Logs	controller_android	extract_logical
SHA-256:	9b738c89fcb1791715e07b7be798e9b1e535d09e4128472f21c2a2e18bfb659e		
SHA-1:	1962b05c55e751ca2d43ae608961cb6f84897582		

log-2017-06-28_6.txt	Flight Logs	controller_android	extract_logical
SHA-256:	e7b3e2e6aac3bf426e2b02b3a5d9f77ba6743b9cff2985eccf4927ac59264fa4		
SHA-1:	3bd5f320731b501ab383a2a1234c2def76f7518c		

log-2017-06-28_7.txt	Flight Logs	controller_android	extract_logical
SHA-256:	bf12729024310bd046e41c00839d38a15411891f90dd314228f577f46c4fe55e		
SHA-1:	6117c72f2ce71f25bb1cb16fe2a283f5eae5f38		

log-2017-06-28_8.txt	Flight Logs	controller_android	extract_logical
SHA-256:	52a71c96665b83e103b3c569dbd2bcbd825859984d067eaa31fdc1df0eeb58a		
SHA-1:	feda577babb85d943b7d81204e7af18eff2272df		

log-2017-06-28_9.txt	Flight Logs	controller_android	extract_logical
SHA-256:	6248344ed0067dc6a968b841977311728f1142160ab8fe0368be8e6bc2848813		
SHA-1:	d87475bc6971b49875679ba8df0c202a4421bdf7		

log-2017-06-29.txt	Flight Logs	controller_android	extract_logical
SHA-256:	40a2379da48c46da1793aec929a1183a005810fd2ee46a94159440fa388a760a		
SHA-1:	4fb75e94e6944066a29937e0f5b12248061c2017		

log-2017-06-29_1.txt	Flight Logs	controller_android	extract_logical
SHA-256:	b3d24f92e600c13c9c042c8e40f1bffb3325beb2aab6563fae1e21021b7e480e		
SHA-1:	952a5cd25a8cf60d122653f1d007ccc14553e566		

log-2017-06-29_10.txt	Flight Logs	controller_android	extract_logical
SHA-256:	1396a317fda9f2f1b613351b5e0c671acedc72245b66eb25259db0904bef6a96		
SHA-1:	46622f454224c0795f421c5fd423af337d7cbeee		

log-2017-06-29_11.txt	Flight Logs	controller_android	extract_logical
SHA-256:	9f3c130ae489a0d25f1ab87a94c425e7f58c372aaf6a28831402e862cc9ea3b7		
SHA-1:	41d8401225e8f40b6936633a1147d14e33855271		

log-2017-06-29_12.txt	Flight Logs	controller_android	extract_logical
SHA-256:	1184cadf8498864caa228d7cabf59d40ceff6ac5883225bf93c6ba444be41d0		
SHA-1:	b57d61a79cac0e62c853202025188373e8cccd5c		

log-2017-06-29_13.txt	Flight Logs	controller_android	extract_logical
SHA-256:	babd988aa485ca8343f1a353277cc6d604edafb47ee03a6ab12166605104f4eb		
SHA-1:	fe57d5a3150922ffe0ec840fc1f1094e70bb584e		

log-2017-06-29_14.txt	Flight Logs	controller_android	extract_logical
SHA-256:	c97f0effc0c777a5668bf0a163c387b87b55cdc18450678015357da1c80ff862		
SHA-1:	b824e454c3327583742d7801f2136bbe85ab3ed7		

log-2017-06-29_2.txt	Flight Logs	controller_android	extract_logical
SHA-256:	d4df915b11bd7a860fe4412593ebdf1d5c48b593b7b89b1b06bfd644183e7859		
SHA-1:	aaef9a3f9c6e6c04dfa5ef3ca565dc2f03087154		

log-2017-06-29_3.txt	Flight Logs	controller_android	extract_logical
SHA-256:	f3ca2c7f7bacdf41ae4b8907ada3f63b2937c7246248517afbad1840992b5cbdb		
SHA-1:	cd3e92e0e318f2ed591eca2aa29c650cdd842820		

log-2017-06-29_4.txt		Flight Logs	controller_android	extract_logical
SHA-256:	ed050910c542f19f8146aa3284fa84ce25cadd2af19a6d447aae9ebf8b092329			
SHA-1:	b2d9336331d59f92383178f3b0848738fb4ac515			

log-2017-06-29_5.txt		Flight Logs	controller_android	extract_logical
SHA-256:	5bac2c0bcc0c207b7d1e73c6a49929596f2dcb32a3a4911c8018d4cecc8278fa			
SHA-1:	7b46470866d8ca43eb6f8cf6300c33a792169857			

log-2017-06-29_6.txt		Flight Logs	controller_android	extract_logical
SHA-256:	bd25814f72c5980ced1f7e7cf27de71642379c2b233fab88e91390c2282d2da2			
SHA-1:	6e6a166b25aba9c958c6e1d8c926c7082e1adfa6			

log-2017-06-29_7.txt		Flight Logs	controller_android	extract_logical
SHA-256:	b9cfe5b879eee1706645800fb90c0be4f786e4b0357f9c0ff377e69849071154			
SHA-1:	228e8a82b1c013b738eedb0bec20451f85616904			

log-2017-06-29_8.txt		Flight Logs	controller_android	extract_logical
SHA-256:	466afa4e871d534f570c0469d12f38017b3ce100971571836551c3caa1aa9817			
SHA-1:	12ce79ccaf8e8225adef085d414cbfaa4e1976d			

log-2017-06-29_9.txt		Flight Logs	controller_android	extract_logical
SHA-256:	2e1ced6f41da03348484203ec456a0ef46c54cb87704725f558a3af47c9e2486			
SHA-1:	277b3b07419a59ef5751eae8d946b2a37971e0f9			

DJIFlightRecord_2017-06-29_[11-15-17].txt		Flight Records	controller_android	extract_logical
SHA-256:	9dd7873f7fbfb2278425ba3f1543db15ea81b0b7c733814f00f025ebd837094			
SHA-1:	2a445770a3d22e4b1fce9bd87e10dd0013255a67			

DJIFlightRecord_2017-06-29_[11-20-32].txt		Flight Records	controller_android	extract_logical
SHA-256:	92a650412e5c516c6a32ac620401b7b7019c7d4339bc6317f0cd619ea4bdf7ae			
SHA-1:	f5fd211ab20a334deef35928fe8de8ded161e98f			

DJI_0002.JPG		Images	drone_sd	extract_physical
SHA-256:	abc58ba30b10e8280e07a11db5163a9874290dd76bfb8124d8c8178bcac72124			
SHA-1:	dce2738fd79c8c9c9bd1e3f68257eeac72f6a7			

DJI_0003.JPG		Images	drone_sd	extract_physical
SHA-256:	a80595fc4dc562dda8355ea5645ba27de8c29cebe7f4b860a9f022c94a878451			
SHA-1:	91ae86314d46b9a814df5486995b46d99fc8e3e3			

DJI_0004.JPG		Images	drone_sd	extract_physical
SHA-256:	d9ad93a5a6b0d831685e27ad41c9342b7e7217198152d851d3e331ee19c510f3			
SHA-1:	b2d0f4f11d687f2dd867c6be5896b2e4acd78c37			

DJI_0006.JPG		Images	drone_sd	extract_physical
SHA-256:	744ab3c8b893b3fdb40cbe3c6ec70756a2f6f78d6dfc94612bd66e9f809f79fa			
SHA-1:	d1770f3dcc6cd2a28161a48a2feb8a911c4baf44			

DJI_0007.JPG		Images	drone_sd	extract_physical
SHA-256:	fe35c4c350c39e33b95fbb8034dc13a7c9f6cc7eb169e451e85499e27251d479			
SHA-1:	74661fca261a4e74b5f7c379ce135af2b2db12b9			

DJI_0008.JPG	Images	drone_sd	extract_physical
SHA-256:	8478e34516836246cdca8cee8bfd13c82ae95b72aa26e80e9c12c4de900fff31		
SHA-1:	7afe816f41819d5508891e68ea35923f4fa20fce		

DJI_0010.JPG	Images	drone_sd	extract_physical
SHA-256:	3148d9fb0072c4522a14bff5cc8a6d8e35850852d56b97114de83cf349b3365a		
SHA-1:	ae8bd84e6b219ffe416a891a17ad4d6ca298a2d0		

DJI_0011.JPG	Images	drone_sd	extract_physical
SHA-256:	2d076163a3d537969bcc9d34fb421a6930061eaf2b1c883eb24c6ad310fe2dd1		
SHA-1:	2e8a9126c020c853990f26fc9cd46ee192682b1b		

DJI_0013.JPG	Images	drone_sd	extract_physical
SHA-256:	6f7b2a40733e00f1c52772fa209dec54350a1b9097dac28d13179a5992bb43b3		
SHA-1:	f443af9bb57ab2d76e8cf8f3a99cb93ea85ec605		

DJI_0014.JPG	Images	drone_sd	extract_physical
SHA-256:	6e5b681afab934878096c909454bf4a257b01d3293bf69abb8adfb48053e74a9		
SHA-1:	f0fc2cb6a796ba55199db3f3a15fa3296b1d51e3		

DJI_0015.JPG	Images	drone_sd	extract_physical
SHA-256:	79d467f059826c93617dc94a8ad5ed32234c3d75adf043cc0cc1868ccc4c8f09		
SHA-1:	6d3f148abc9f1926347c387e9a66346aa4a14c77		

DJI_0016.JPG	Images	drone_sd	extract_physical
SHA-256:	a6c877233493fd603062ed737615fee8a8ceb55ba2b6da09e5b19a232e0b76d9		
SHA-1:	a1ab9fc323ef5a77870c1b5bea8df1c72ffc9344		

DJI_0017.JPG	Images	drone_sd	extract_physical
SHA-256:	eee797d80b320a4328022be386c0d9c825c3eb18d86292a069abac1e7da14b58		
SHA-1:	972d904cd04e16d6392df4f5585d696744d544f7		

DJI_0019.JPG	Images	drone_sd	extract_physical
SHA-256:	5cd0b44634eb69338cc27858c94f286628ff58784f59ff3dc2307adec903c8ed		
SHA-1:	4238e64fa022c2e0a91a4c2fcd8438d00d907fac		

DJI_0020.JPG	Images	drone_sd	extract_physical
SHA-256:	7df85c310d22165acadfb8c9ce832761b368383f34ea42dae50bbcd011c8a2ea		
SHA-1:	1d42c6d0b52da2eed98c55b49fbb8205406eda63		

2017_06_29_11_20_42.info	Videos	controller_android	extract_logical
SHA-256:	9a5661921fe118d55c89cfbd0abd9f26d668d5d3208f4c4437fe1a32551f9448		
SHA-1:	2518bb9cfd18d34e91d554ae61c166c524a1c99f		

2017_06_29_11_20_42.mp4	Videos	controller_android	extract_logical
SHA-256:	d39d349f9f2ea9826b31eb50a933b7b26f2a915c292a5ceaea56be7da9eaa428		
SHA-1:	117cd6d472012dc8aa1c5d5b5c960e0f00c157e4		

DJI_0001.MOV	Videos	drone_sd	extract_physical
SHA-256:	83d34e5880b8be3bde56bd87afebd103592a5e5994304a995d036034962a8272		
SHA-1:	19fb83d04583cf8ecddac75ca45229900afe9bb7		

DJI_0001_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	7b8d41c1775714af2eadff7cf56c9acfbf4b0d80a1c53867d061be7e25eea680			
SHA-1:	605ca2b769e5b9892d7bf41b4b89f01e4e3c855d			

DJI_0005.MOV		Videos	drone_sd	extract_physical
SHA-256:	624ccfe3d1f63aad44f49156a8f89326e679a81087999ebb77f6707815e350bb			
SHA-1:	8d7da3ee328562929d5a657b0b7292f349e84a90			

DJI_0005_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	84b82a66fa36d126ed580f2d91bc47d6f4787b93bec75585ba23dfefb4068eb19			
SHA-1:	8e9da7f5e2f24d861c5a46845cbf0b03aa4b4a0c			

DJI_0009.MOV		Videos	drone_sd	extract_physical
SHA-256:	d1c7d4c9532dd338776e728447b3f9ac9637aa0df2403597cebae87a12fce7ae			
SHA-1:	6083d7e5f675e1f216ef82fa9814f4411e40634f			

DJI_0009_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	07193b3b3f1f024c7b8e66cbe104bd4106db3269a6f278245056cb63ec837132			
SHA-1:	e78d17014597433209d68326585d6ec003af7818			

DJI_0012.MOV		Videos	drone_sd	extract_physical
SHA-256:	14fbc8239759ab3953aca18c3b7ef9beccc96ee09c67943bfe7483dbfff99068			
SHA-1:	a54f6de5eef35eba6ca50cc376aefb3190d068b2			

DJI_0012_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	c29a9498a07292ad1a133d77584e928ee5a05147cfd4d3b770f4d7c6ada17406			
SHA-1:	05c79c4913892219754bb52c62e2a948eb719cfd			

DJI_0018.MOV		Videos	drone_sd	extract_physical
SHA-256:	bb2ec42222e8a87726519e1846155bc5937aa0b31f6a797cd97e71376e65aa45			
SHA-1:	8d1f7ae484a6b6821af19b52ede60e7df313f403			

DJI_0018_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	7eecdb89bdc561442c19cd3b6c462e508398bb945657f0acb4fb3a3b29b18f55			
SHA-1:	47f4b85e8c71ec7a362b0772e9fda9d7f4acd512			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	2	2	0
Databases	7	0	0
Flight Logs	35	0	0
Flight Records	2	2	2
Videos	2	2	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	15	15	0
Videos	10	10	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-17 12:22:46 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df005_external.001

2026-06-17 12:22:46 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df005_external.001

2026-06-17 12:23:39 parser_android v- rc=n/a

cmd: DF005_flight.zip C:\Users\Floris\Documents\dfdof_output\CASE-df005-2017-android\p2_image_parsing\controller_android_parsed
out: controller_android_parsed

2026-06-17 12:24:00 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797701
out: DJI_0001.MOV

2026-06-17 12:24:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797702
out: DJI_0002.JPG

2026-06-17 12:24:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797703
out: DJI_0003.JPG

2026-06-17 12:24:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797704
out: DJI_0004.JPG

2026-06-17 12:24:24 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797705
out: DJI_0005.MOV

2026-06-17 12:24:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797706
out: DJI_0006.JPG

2026-06-17 12:24:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797707
out: DJI_0007.JPG

2026-06-17 12:24:37 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797708
out: DJI_0008.JPG

2026-06-17 12:24:48 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797709
out: DJI_0009.MOV

2026-06-17 12:24:58 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797710
out: DJI_0010.JPG

2026-06-17 12:24:58 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797711
out: DJI_0011.JPG

2026-06-17 12:25:06 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797712
out: DJI_0012.MOV

2026-06-17 12:25:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797713

out: DJI_0013.JPG

2026-06-17 12:25:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797714

out: DJI_0014.JPG

2026-06-17 12:25:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797715

out: DJI_0015.JPG

2026-06-17 12:25:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797716

out: DJI_0016.JPG

2026-06-17 12:25:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797717

out: DJI_0017.JPG

2026-06-17 12:25:17 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797718

out: DJI_0018.MOV

2026-06-17 12:25:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797719

out: DJI_0019.JPG

2026-06-17 12:25:19 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797720

out: DJI_0020.JPG

2026-06-17 12:25:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798725

out: DJI_0001_1.MOV

2026-06-17 12:25:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798726

out: DJI_0005_1.MOV

2026-06-17 12:25:25 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798727

out: DJI_0009_1.MOV

2026-06-17 12:25:26 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798728

out: DJI_0012_1.MOV

2026-06-17 12:25:27 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798729

out: DJI_0018_1.MOV

2026-06-17 12:25:28 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-06-29_[11-15-17].txt DJIFlightRecord_2017-06-29_[11-15-17].csv

out: DJIFlightRecord_2017-06-29_[11-15-17].csv

2026-06-17 12:25:29 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-06-29_[11-20-32].txt DJIFlightRecord_2017-06-29_[11-20-32].csv

out: DJIFlightRecord_2017-06-29_[11-20-32].csv

2026-06-17 12:25:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_11_20_42.mp4

out: 2017_06_29_11_20_42_exif.json

2026-06-17 12:25:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG
out: DJI_0002_exif.json

2026-06-17 12:25:32 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.JPG
out: DJI_0003_exif.json

2026-06-17 12:25:32 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.JPG
out: DJI_0004_exif.json

2026-06-17 12:25:32 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.JPG
out: DJI_0006_exif.json

2026-06-17 12:25:32 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0007.JPG
out: DJI_0007_exif.json

2026-06-17 12:25:33 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0008.JPG
out: DJI_0008_exif.json

2026-06-17 12:25:33 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0010.JPG
out: DJI_0010_exif.json

2026-06-17 12:25:33 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0011.JPG
out: DJI_0011_exif.json

2026-06-17 12:25:34 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0013.JPG
out: DJI_0013_exif.json

2026-06-17 12:25:34 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0014.JPG
out: DJI_0014_exif.json

2026-06-17 12:25:34 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0015.JPG
out: DJI_0015_exif.json

2026-06-17 12:25:34 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0016.JPG
out: DJI_0016_exif.json

2026-06-17 12:25:35 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0017.JPG
out: DJI_0017_exif.json

2026-06-17 12:25:35 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0019.JPG
out: DJI_0019_exif.json

2026-06-17 12:25:35 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0020.JPG
out: DJI_0020_exif.json

2026-06-17 12:25:36 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MOV
out: DJI_0001_exif.json

2026-06-17 12:25:36 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001_1.MOV

out: DJI_0001_1_exif.json

2026-06-17 12:25:37 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MOV

out: DJI_0005_exif.json

2026-06-17 12:25:37 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005_1.MOV

out: DJI_0005_1_exif.json

2026-06-17 12:25:38 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009.MOV

out: DJI_0009_exif.json

2026-06-17 12:25:39 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009_1.MOV

out: DJI_0009_1_exif.json

2026-06-17 12:25:39 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0012.MOV

out: DJI_0012_exif.json

2026-06-17 12:25:40 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0012_1.MOV

out: DJI_0012_1_exif.json

2026-06-17 12:25:40 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0018.MOV

out: DJI_0018_exif.json

2026-06-17 12:25:40 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0018_1.MOV

out: DJI_0018_1_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.